

24.1 Ultrasound & X-rays

Question Paper

Course	CIE A Level Physics
Section	24. Medical Physics
Topic	24.1 Ultrasound & X-rays
Difficulty	Easy

Time allowed: 60

Score: /46

Percentage: /100

Question 1a

State what is meant by the specific acoustic impedance of a medium.

[2 marks]

Question 1b

Table 1.1 shows the density and speed of ultrasound in soft tissue and in air.

Table 1.1

	density / kg m^{-3}	speed of ultrasound / m s^{-1}
air	1.29	330
soft tissue	1060	1540
bone	1600	4000

Using the data in Table 1.1, calculate

(i)

the acoustic impedance of air,

[1]

(ii)

the acoustic impedance of soft tissue,

[1]

(iii)

the acoustic impedance of bone.

[1]

[3 marks]

Question 1c

The intensity reflection coefficient α for two media is given by

$$\alpha = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$$

Where Z_1 and Z_2 are the specific acoustic impedances of the two media.

Explain how this equation can be used to predict if ultrasound will be reflected or transmitted at the boundary between two materials.

[3 marks]

Question 1d

State what is meant by **impedance matching** and explain how it may be achieved.

[2 marks]

Question 2a

Fig. 1.1 shows an incomplete diagram of a piezoelectric transducer.

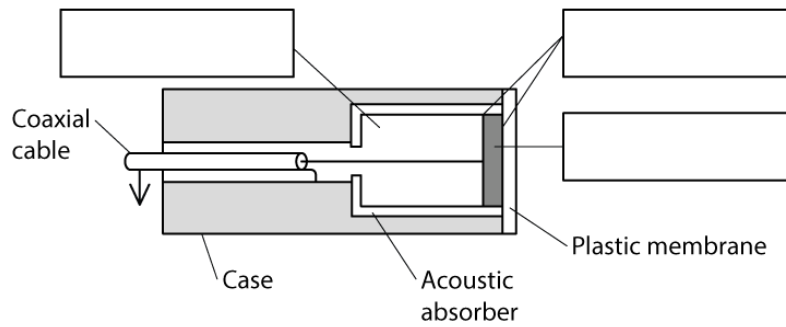


Fig. 1.1

(i)

Fill in the three missing labels on Fig. 1.1.

[3]

(ii)

State the purpose of the components in **(a)(i)** in relation to the generation of ultrasound waves.

[3]

[6 marks]

Question 2b

(i)

State **two** main pieces of information about internal body structures that ultrasound can provide.

[2]

(ii)

State the properties of the ultrasound signal that provide these pieces of information.

[2]

[4 marks]**Question 2c**

(i)

Explain what is meant by ionising radiation and why it can be harmful.

[2]

(ii)

Place ticks (✓) next to the scans in Table 1.1 which use ionising radiation.

Table 1.1

Type of scan	
X-ray Imaging	
Ultrasound	
PET scan	
CT scan	

[2]

[4 marks]

Question 2d

Describe the differences between CT scanning and X-ray imaging.

[3 marks]

Question 3a

Fig. 1.1 shows a simplified diagram of an X-ray tube.

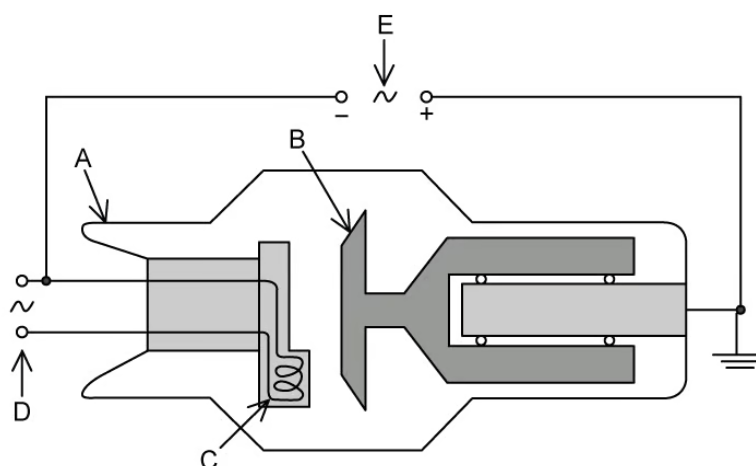


Fig. 1.1

State the name and purpose of

- | | | |
|-------|-------------|-------------------|
| (i) | Component A | [2] |
| (ii) | Component B | [2] |
| (iii) | Component C | [2] |
| (iv) | Component D | [2] |
| (v) | Component E | [2] |
| | | [10 marks] |

Question 3b

The accelerating potential difference between the cathode and the anode of an X-ray tube is 75 kV.

(i)

State the maximum energy, in eV, of the photons produced in the X-ray tube.

[1]

(ii)

Calculate the minimum wavelength of photons in the X-ray beam.

[3]

[4 marks]

Question 3c

A parallel beam of X-rays is incident on a medium with a linear absorption coefficient μ .

State what is meant by the linear absorption coefficient.

[2 marks]

Question 3d

Table 1.1 shows the linear absorption coefficient μ for an X-ray beam in blood and in muscle.

Table 1.1

	μ / cm^{-1}
blood	0.23
muscle	0.22

(i)

State what is meant by the contrast of an X-ray.

[1]

(ii)

Using the data in Table 1.1, explain why an X-ray image of blood vessels in muscle would have poor contrast.

[2]

[3 marks]



Model Answers: Easy

1a

Specific acoustic impedance is defined as:

EITHER

- The product of the density (of medium) and speed (of ultrasound); [1 mark]
- In the medium; [1 mark]

OR

- $Z = \rho c$ [1 mark]
- Where ρ is the density (of the medium) and c is the speed (of the ultrasound) in the medium; [1 mark]

[Total: 2 marks]

If you're stating an equation for a definition, make sure you state what **each variable** means to get full marks.

1b

Calculate the acoustic impedance of...

(i) Air:

- $Z = \rho c$
- $Z_{\text{air}} = 1.29 \times 330$
- $Z_{\text{air}} = 426 \text{ kg m}^{-2} \text{ s}^{-1}$ [1 mark]

(ii) Soft tissue:

- $Z = \rho c$
- $Z_{\text{tissue}} = 1060 \times 1540$

- $Z_{\text{tissue}} = 1.63 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$ [1 mark]

(iii) Bone:

- $Z = \rho c$
- $Z_{\text{bone}} = 1600 \times 4000$
- $Z_{\text{bone}} = 6.4 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$ [1 mark]

[Total: 3 marks]

Always state the appropriate unit for your final answer if the question does include it already.

Here, the units are $[\text{kg}] [\text{m}^{-3}] \times [\text{m}] [\text{s}^{-1}] = [\text{kg}] [\text{m}^{-2}] [\text{s}^{-1}]$. You **must** be familiar with the law of indices in A-level physics.

1c

The equation can be used to predict if ultrasound will be reflected or transmitted at the boundary between two materials by:

- Calculate the difference between the values of Z / acoustic impedance (of the two materials); [1 mark]
- If the values of Z are similar $\alpha \approx 1$ there will be complete transmission **OR** little / no reflection; [1 mark]
- If the values of Z are different $\alpha \approx 0$ there will be little / no transmission **OR** large / complete reflection; [1 mark]

[Total: 3 marks]

1d

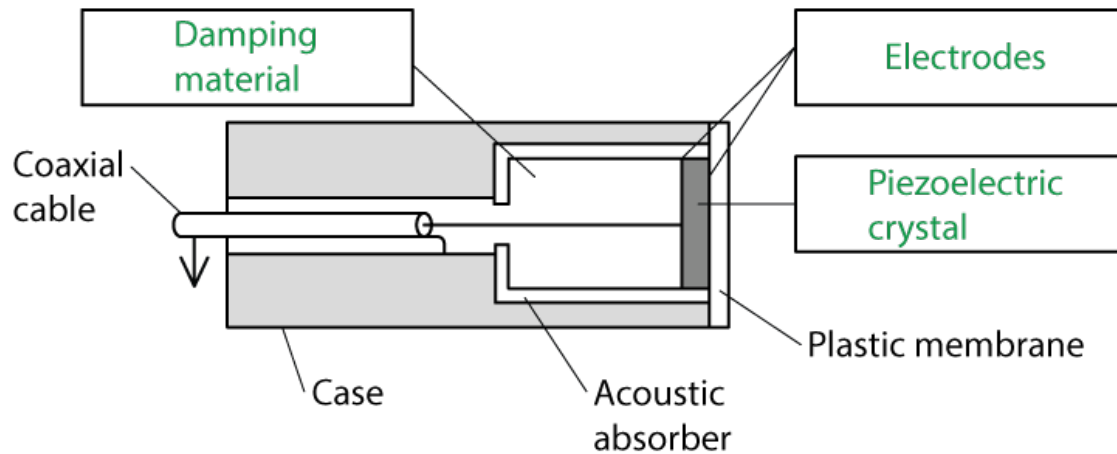
Impedance matching is:

- When a coupling gel is used that has the same / matching acoustic impedance as the skin / body; [1 mark]
- Which reduces / removes reflection at the skin; [1 mark]

[Total: 2 marks]

2a

(i) The correct labels are as follows:



- *One mark for each correct label*

(ii) The purpose of each component is:

- Piezoelectric crystal: produces ultrasound waves from vibrations / resonance when an alternating p.d. is applied across it; [1 mark]
- Electrodes: to apply an alternating p.d. across the (piezoelectric) crystal; [1 mark]
- Backing / damping material: to damp the oscillations **OR** to stop the crystal from vibrating too much; [1 mark]

[Total: 6 marks]

2b

(i) Ultrasound can provide information about the:

- Depth of a boundary; [1 mark]
- Nature of a boundary; [1 mark]

(ii) The properties of the ultrasound signal are:

- The time delay / time between transmission and receipt (of the pulse) gives information about the depth (of the boundary); [1 mark]
- The intensity of the reflection gives information about the nature (of the boundary); [1 mark]

[Total: 4 marks]

2c

(i) Ionising radiation is:

- Radiation that can cause ionisation of atoms / cells (or of any other material it passes through); [1 mark]
- That is highly energetic so can cause damage to DNA / cause cell mutation; [1 mark]

(ii) The scans that use ionising radiation are:

Type of scan	
X-ray Imaging	✓
Ultrasound	
PET scan	✓
CT scan	✓

- Any two correct; [1 mark]
- All three correct; [1 mark]

[Total: 4 marks]

2d

The differences between CT scanning and X-ray imaging are:

- An X-ray image is a single 'shadow' image / 2-D image (of the whole structure); [1 mark]
- A CT image is made up of many X-rays images of a (thin) slice (through a structure); [1 mark]
- At different angles / combined to make a 3-D image; [1 mark]

[Total: 3 marks]

X-ray Imaging	✓
Ultrasound	
PET scan	✓
CT scan	✓

- Any two correct; [1 mark]
- All three correct; [1 mark]

[Total: 4 marks]

2d

The differences between CT scanning and X-ray imaging are:

- An X-ray image is a single 'shadow' image / 2-D image (of the whole structure); [1 mark]
- A CT image is made up of many X-rays images of a (thin) slice (through a structure); [1 mark]
- At different angles / combined to make a 3-D image; [1 mark]

[Total: 3 marks]

3a

The name and purpose of...

(i) Component A:

- Name: evacuated (glass) tube; [1 mark]
- Purpose: so the electrons do not collide with any particles on their way to the metal target; [1 mark]

(ii) Component B:

- Name: anode / (metal) target; [1 mark]
- Purpose: to produce X-rays when accelerated electrons hit the anode / target; [1 mark]

(iii) Component C:

- Name: cathode / (heated) filament; [1 mark]

- Purpose: to release electrons from the surface of the cathode by thermionic emission; [1 mark]

(iv) Component D:

- Name: low voltage supply; [1 mark]
- Purpose: to heat the filament; [1 mark]

(v) Component E:

- Name: high voltage supply; [1 mark]
- Purpose: to create a large potential difference between the cathode and the anode; [1 mark]

[Total: 10 marks]

3b

(i) The maximum energy of the X-rays is:

- Energy gained by electron = energy of photon
- (Hence) maximum photon energy = 75 keV [1 mark]

(ii) Calculate the minimum wavelength of photons in the X-ray beam:

List the known quantities:

- Planck constant, $h = 6.63 \times 10^{-34} \text{ J s}$
- Charge on the electron, $e = 1.60 \times 10^{-19} \text{ C}$
- Speed of light, $c = 3.00 \times 10^8 \text{ m s}^{-1}$

Calculate the minimum wavelength using the photon energy equation:

- Energy of a photon: $E = \frac{hc}{\lambda}$ [1 mark]

- $eV = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{eV}$

- $\lambda = \frac{(6.63 \times 10^{-34}) \times (3 \times 10^8)}{(1.6 \times 10^{-19}) \times (75 \times 10^3)}$ [1 mark]
- $\lambda = 1.66 \times 10^{-11} \text{ m}$ [1 mark]

[Total: 4 marks]

Work done on a charge, $W = QV$ where Q is the charge and V the potential difference. Since the charge of an electron is e this means that $W = E$ (energy transferred) = eV .

3c

The linear absorption coefficient is:

- $I = I_0 e^{-\mu x}$ [1 mark]
- Where I_0 is the intensity of the incident beam, I is the intensity of the reflected beam and x = distance travelled through the material; [1 mark]

[Total: 2 marks]

Make sure not just to state the equation, but explain each of the variables for full marks. You must be detailed in some equations e.g. just writing 'I is the intensity' will not get you the mark, because **two** beams are involved, so you must state the intensity of which one this is referring to.

3d

(i) Contrast is defined as:

- The difference in the degree of blackening (of regions of the image); [1 mark]

(ii) The image would have poor contrast because:

- The values of μ (of blood and muscle) are (very) similar; [1 mark]
- So they absorb a similar amount of X-rays (for the same thickness) (hence poor contrast); [1 mark]

[Total: 3 marks]